AIORI Hackathon 3

Problem Statement PS-003

A3-PS003-TC293 Federated Traffic-Flow Prediction

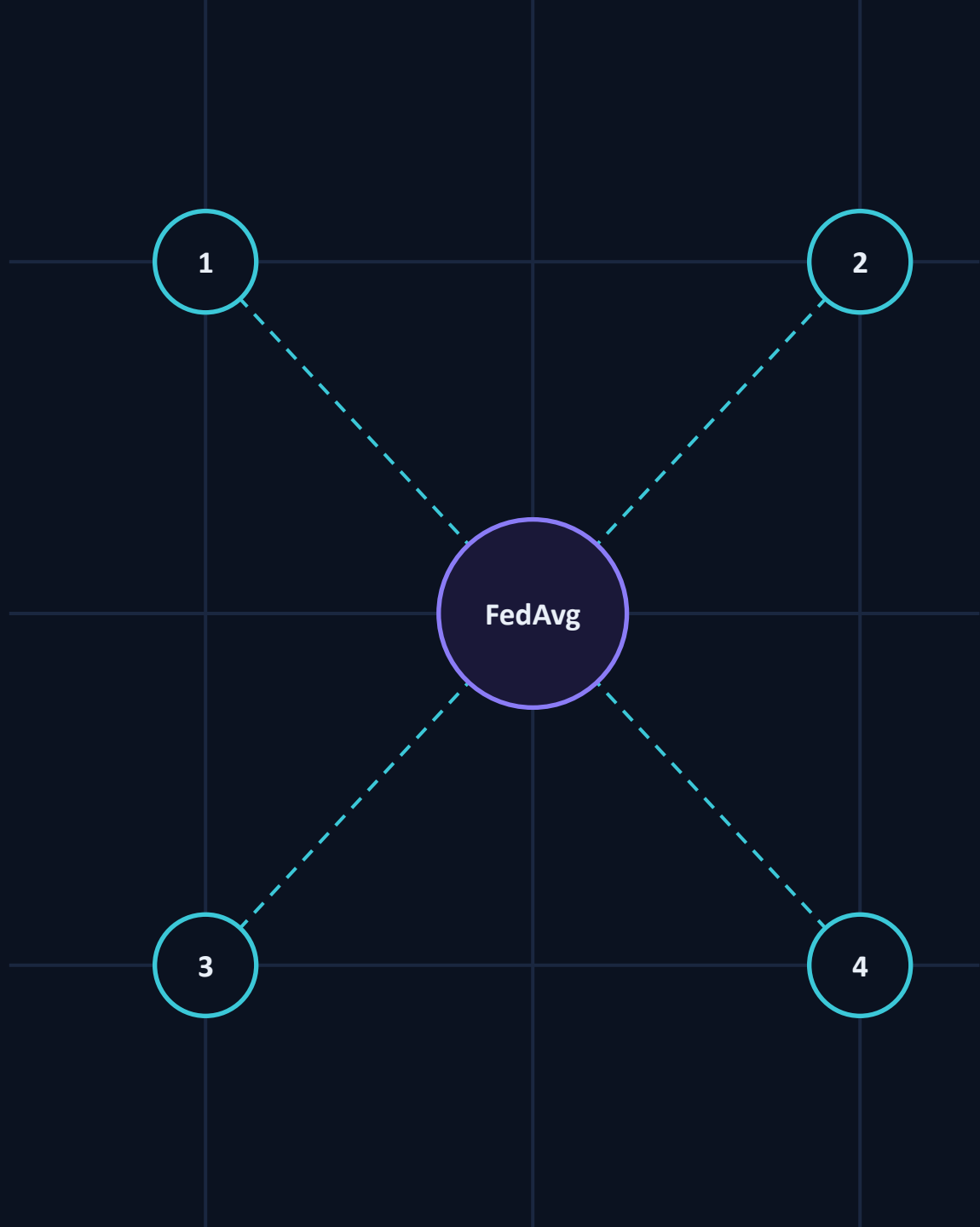
Privacy-Preserving and Communication-Efficient Edge Intelligence

Team: INVARIANT

Members: MD Nayab Ahmad · Shaurya · Dr Sumit Raj

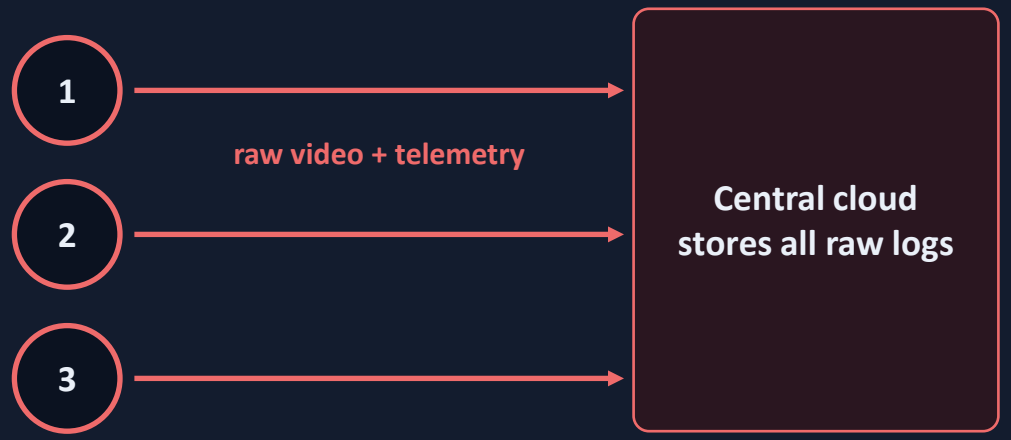
Proposal and experimental plan — results will be reported after the experiments are run.

October 13, 2026



Centralized forecasting is costly and exposed

CONVENTIONAL: CENTRALIZED LEARNING



High cellular bandwidth; concentrated private travel logs.

PROPOSED: FEDERATED LEARNING



Raw data stays local; only model parameters travel.

Problem definition

Centralized traffic forecasting needs edge cameras and sensors to stream raw data continuously, consuming bandwidth and concentrating private travel logs.

Project objective

Build a federated simulator and compare centralized vs FedAvg forecasting: MAE / RMSE against bytes per round, IID vs Non-IID, across 3 seeds.

Learn together, keep the data local

1

Train locally

Each simulated intersection trains a small forecasting model on its own historical traffic data.

2

Share only weights

Clients send model parameters (updates) to the server — not their raw traffic records.

3

Average and redistribute

The server combines updates with FedAvg and sends the improved global model back.

Core algorithm (existing)

- Federated Averaging (FedAvg), McMahan et al., AISTATS 2017
- Clients run local SGD; the server takes a weighted average of their models
- Built for private data and limited communication

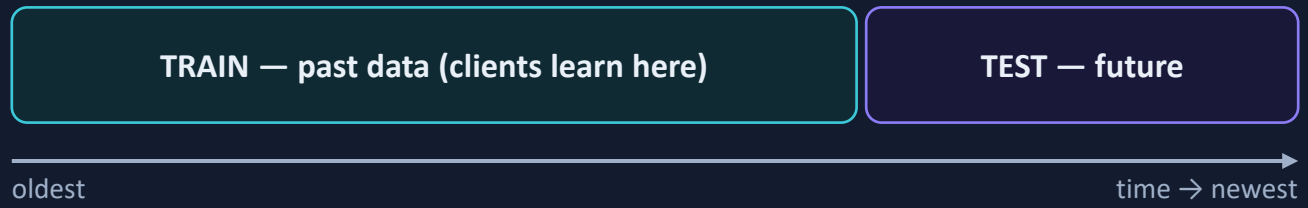
Our contribution (proposed)

- Traffic-flow forecasting on public historical data
- Centralized vs FedAvg under the same split, target and test set
- IID vs Non-IID client partitions
- MAE / RMSE vs bytes per round, plus convergence, over 3 seeds

Data pipeline and experimental setup

1. **Public historical dataset**
   Kaggle / UCI traffic data.
   Final dataset: TO BE FINALIZED
2. **Preprocess**
   Clean, handle gaps, scale (fit on train only), build lag windows.
3. **Chronological split**
   Earlier period → train. Later period → test. No shuffling across the boundary.
4. **Partition to clients**
   3–5 virtual intersections, each holding its own training slice.
5. **IID / Non-IID scenarios**
   Same model, metrics and seeds in both settings.

Chronological split — preventing temporal leakage



Test period is never seen in training; split ratio is set once the dataset is final.

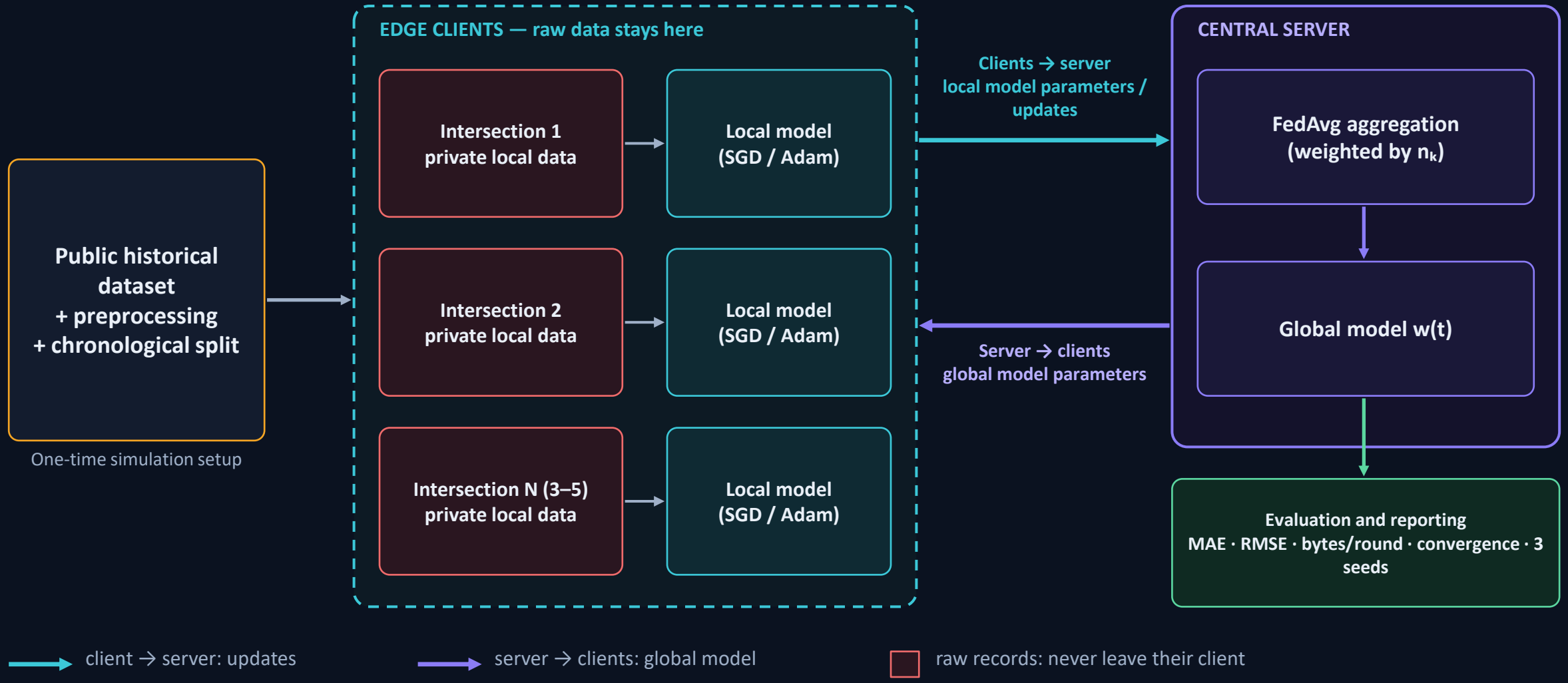
Client partitioning

IID: clients get statistically similar slices of the training period.

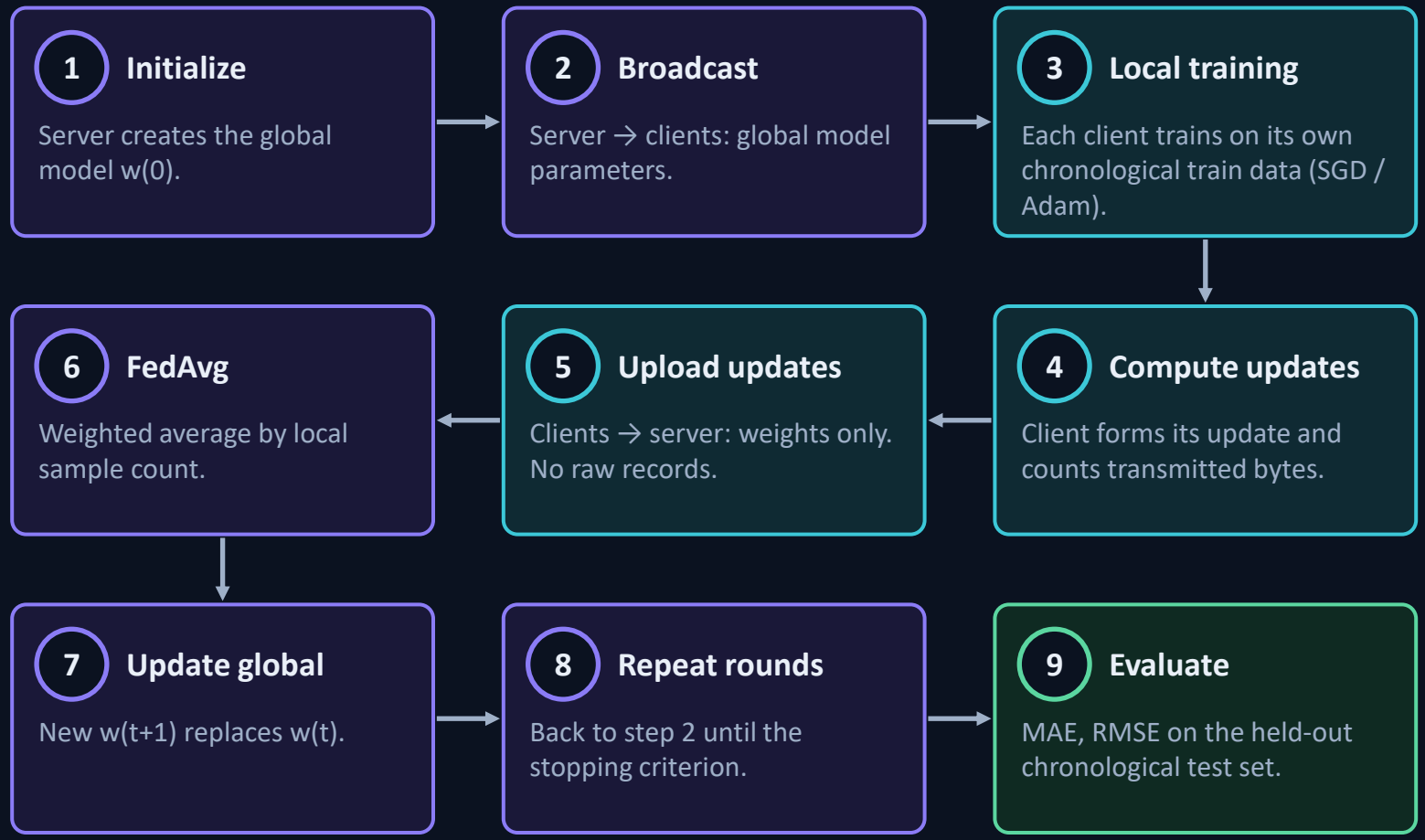
Non-IID: clients differ, e.g. different peak-hour patterns.

Train slices only; the held-out test set stays chronological.

Solution architecture and design



FedAvg data flow, round by round



Weighted FedAvg

$$w_{t+1} = \Sigma_k \left(n_k / \Sigma_j n_j \right) \cdot w_k(t+1)$$

n_k = local training examples at client k

$w_k(t+1)$ = client k's updated parameters

RAW RECORDS — stay on the client

MODEL UPDATES — sent to the server

Every round, clients start from the same global model. Local epochs, batch size, optimizer and aggregation must match the final implementation.

Model design and proposed optimizations



Planned efficiency levers (all proposed)

- Lightweight model → small update payload
- Local training before each upload
- Log bytes per round
- Tune local epochs / batch size

Aspect	Conventional (centralized)	Proposed (federated)
Raw traffic data	Streamed to a central cloud	Stays on each simulated client
Uplink payload	Raw video / telemetry	Model parameters / updates only
Privacy exposure	Central store of travel logs	Reduced; no formal guarantee (no DP / secure aggregation)
Bandwidth	High, continuous	Target ~80%+ reduction in transmitted bytes — unverified, to be measured
Evidence	Centralized baseline: planned	FedAvg runs: IID / Non-IID × 3 seeds, planned

Evaluation framework and evidence plan

Metric	What it tells judges	How measured
MAE	Average absolute forecast error; lower is better	Held-out chronological test set
RMSE	Penalizes large misses more; lower is better	Same test set and target
Comm. bytes	Cost of model traffic per round and in total	Logged each round, then summed
Convergence	How loss falls over communication rounds	Loss vs round, IID vs Non-IID
Variance	Stability, not a lucky run	3 distinct seeds; mean ± std

Is the trade-off useful?

- Compare FedAvg and centralized on the same split, target and test set
- Accept FedAvg if MAE / RMSE stay within a pre-agreed margin while bytes drop
- Report the IID vs Non-IID gap and the spread across seeds

The ~80%+ byte reduction in the proposal is a target to verify, not a result.

Roadmap, expected impact, and limitations

ROADMAP — all milestones planned, no calendar dates



Expected benefits

- Less need to centralize raw traffic data
- Potentially lower communication cost — to be verified
- Evidence on behavior under heterogeneous traffic
- Reproducible evaluation on historical data

Scope and limitations

- Simulation, not a live city deployment
- Historical data, no live camera feeds
- No formal privacy guarantee (no differential privacy or secure aggregation)
- Accuracy and bandwidth gains must be shown by measurement

Conclusion and references

Learn collaboratively. Keep raw traffic data local. **Measure the communication cost.**

Problem

Centralized forecasting streams raw data: bandwidth cost and private logs.

Approach

Simulated intersections train locally; FedAvg merges weight updates.

Evaluation

MAE, RMSE, bytes per round, convergence; IID vs Non-IID; 3 seeds.

Expected contribution

Measured accuracy vs communication trade-off on traffic data.

References

[1] AIORI Hackathon 3. PS-003: Federated Traffic-Flow Prediction with Communication-Cost Evaluation (official problem statement).

[2] McMahan, Moore, Ramage, Hampson, Agüera y Arcas (2017). Communication-Efficient Learning of Deep Networks from Decentralised Data. AISTATS, PMLR 54. *(https://proceedings.mlr.press/v54/mcmahan17a.html)*

[3] Bangalore Traffic Dataset *(https://www.kaggle.com/datasets/preethamgouda/banglore-city-traffic-dataset)*

THANK YOU FOR YOUR ATTENTION!